Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **full-stack products** using **React/TypeScript** and **Python services**, shipping reliable, scalable workflows with strong product collaboration and AI-assisted development practices.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving reliability, observability, and maintainability across production full-stack systems.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **production stability** and accelerating issue triage across cross-functional product, engineering, and data stakeholders.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, translating product requirements into polished interfaces and secure backend APIs while improving end-to-end UX responsiveness and workflow clarity.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through maintainable tests, incremental shipping, and strong code-review collaboration loops.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling reproducible behavior analysis and evaluation workflows that informed product decisions, technical strategy, and scalable implementation priorities.
- Produced technical summaries and debugging diagnostics that converted findings into implementation plans, improving **cross-functional collaboration** and reducing ambiguity across design, research, and engineering delivery cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across distributed platform services.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** via automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving service reliability and user-facing responsiveness in high-throughput collaborative product operations.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through scripted validation, robust rollback, and production-quality checks.

Skills

Languages: TypeScript, JavaScript, Python, Java, C++, SQL, Bash
Frontend: React, TypeScript, JavaScript, Responsive UI, Product-Focused UX Workflows
Backend & Platform: Flask, FastAPI, REST APIs, Microservices, WebSockets, Distributed Services
Cloud & Reliability: AWS (S3, Glue, ECS), Docker, Kubernetes, Azure, Linux/UNIX, Monitoring
Engineering Practices: CI/CD, Unit/Integration Testing, Code Reviews, Git, Jenkins, Agile/Scrum

Projects

PiXMod | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI visual editing platform using **SAM** segmentation and **Stable Diffusion** generation, processing **1K+ assets** while improving user-facing interaction quality, metadata indexing, and end-to-end workflow reliability.

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*

- Built a scalable fintech product supporting **50K+ users** and **1M+ transactions** with secure integrations, real-time updates, and **Kubernetes**-backed deployments centered on performance, reliability, and strong user experience.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-assisted SQL workspace supporting **10+ databases**, improving query authoring and schema exploration through robust full-stack service interfaces, repeatable validations, and developer-focused usability improvements.